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gayathri@Gayathri: ~/OSSP/E x + v
gayathri@Gayathri:~$ mkdir -p ~/OSSP/Experiment3
gayathri@Gayathri:~$ cd ~/OSSP/Experiment3
gayathri@Gayathri:~/OSSP/Experiment3$ pwd
/home/gayathri/OSSP/Experiment3
gayathri@Gayathri:~/OSSP/Experiment3$ gedit process.c

** (gedit:543): WARNING **: 08:43:30.429: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:543): WARNING **: 08:43:30.429: Could not load Peas repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

Open process.c
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 #include <sys/types.h>
5 #include <sys/wait.h>
6
7 int main()
8 {
9     pid_t pid;
10
11     printf("Parent process started.\n");
12     printf("Parent PID : %d\n", getpid());
13     printf("Parent PPID : %d\n", getppid());
14
15     pid = fork();
16
17     if (pid < 0)
18     {
19         perror("fork failed");
20         return 1;
21     }
22
23     if (pid == 0)
24     {
25         // Child process
26         printf("\n--- CHILD PROCESS ---\n");
27         printf("Child PID : %d\n", getpid());
28         printf("Child PPID : %d\n", getppid());
29
30         printf("Child is now running.\n");
31
32         sleep(10);
33
34         printf("Child is terminating.\n");
35         exit(0);
36     }
37     else
38     {
39         // Parent process
40         printf("\n--- PARENT PROCESS ---\n");
41         printf("Parent PID : %d\n", getpid());
42         printf("Child PID : %d\n", pid);
43
44         printf("Parent is waiting for child...\n");
45
46         wait(NULL);
47
48         printf("Child has terminated.\n");
49         printf("Parent process is now terminating.\n");
50     }
51
52     return 0;
53 }
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gayathri@Gayathri:~/OSSP/Experiment3$ gcc process.c -o process
gayathri@Gayathri:~/OSSP/Experiment3$ ls
process process.c
gayathri@Gayathri:~/OSSP/Experiment3$ ./process
Parent process started.
Parent PID : 5561
Parent PPID : 334

--- PARENT PROCESS ---
Parent PID : 5561
Child PID : 5562
Parent is waiting for child...

--- CHILD PROCESS ---
Child PID : 5562
Child PPID : 5561
Child is now running.
Child is terminating.
Child has terminated.
Parent process is now terminating.
gayathri@Gayathri:~/OSSP/Experiment3$ ./process
Parent process started.
Parent PID : 5572
Parent PPID : 334

--- PARENT PROCESS ---
Parent PID : 5572
Child PID : 5573
Parent is waiting for child...

--- CHILD PROCESS ---
Child PID : 5573
Child PPID : 5572
Child is now running.
Child is terminating.
Child has terminated.
Parent process is now terminating.
gayathri@Gayathri:~/OSSP/Experiment3$ tps -ef | grep process
Command 'tps' not found, did you mean:
  command 'pps' from deb libpmix-bin (6.0.0+really5.0.9-3build1)
  command 'ips' from deb ips (4.2-1)
  command 'ps' from deb procs (2:4.0.4-9ubuntu1)
  command 'jps' from deb openjdk-25-jdk-headless (25.0.3+9-2~26.04.2)
  command 'jps' from deb openjdk-11-jdk-headless (11.0.31+11-1ubuntu1~26.04.2)
  command 'jps' from deb openjdk-17-crack-jdk-headless (17.0.19+10-0ubuntu1~26.04.1)
  command 'jps' from deb openjdk-17-jdk-headless (17.0.19+10-1~26.04.2)
  command 'jps' from deb openjdk-21-crack-jdk-headless (21.0.11+10-0ubuntu1~26.04.1)
  command 'jps' from deb openjdk-21-jdk-headless (21.0.11+10-1~26.04.2)
  command 'jps' from deb openjdk-25-crack-jdk-headless (25.0.3+9-0ubuntu1~26.04.1)
  command 'jps' from deb openjdk-26-jdk-headless (26.0.1+8-2~26.04.2)
  command 'jps' from deb openjdk-8-jdk-headless (8u492-ga-us2-0ubuntu1~26.04.1)
  command 'qps' from deb qps (2.12.0-0ubuntu1)
  command 'bps' from deb bpfcc-introspection (0.35.0+ds-1ubuntu2)
Try: sudo apt install <deb name>
gayathri@Gayathri:~/OSSP/Experiment3$ ps -o pid,ppid,state,stat,cmd -C process
  PID  PPID  S  STAT  CMD
gayathri@Gayathri:~/OSSP/Experiment3$ top
top - 08:48:15 up 3 min, 1 user, load average: 0.05, 0.06, 0.02
Tasks: 35 total, 1 running, 34 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
```

```
gayathri@gayathri: ~/OSSP/E x + v
top - 08:48:24 up 3 min, 1 user, load average: 0.04, 0.06, 0.02
Tasks: 35 total, 1 running, 34 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.1 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 15786.6 total, 14270.2 free, 671.0 used, 1044.7 buff/cache
MiB Swap: 4096.0 total, 4096.0 free, 0.0 used, 15115.6 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0   24124   15388 11584 S   0.0   0.1   0:00.69 systemd
    2 root        20   0    3180    2200   2072 S   0.0   0.0   0:00.00 init-systemd(Ub
    8 root        20   0    3180    2044    1940 S   0.0   0.0   0:00.00 init
   48 root       19  -1   42204   16808   15524 S   0.0   0.1   0:00.17 systemd-journal
   80 systemd+    20   0   22540   14564   11940 S   0.0   0.1   0:00.07 systemd-resolve
   87 root        20   0   35472   12456    9252 S   0.0   0.1   0:00.44 systemd-udev
  109 root        20   0    2888    1948    1820 S   0.0   0.0   0:00.05 chronyd-starter
  110 root        20   0    4472    3088    2840 S   0.0   0.0   0:00.01 cron
  111 message+    20   0    8816    5452    4764 S   0.0   0.0   0:00.07 dbus-daemon
  126 root        20   0   44092   29804   14660 S   0.0   0.2   0:00.29 networkd-dispat
  162 root        20   0   18436    9384    8140 S   0.0   0.1   0:00.12 systemd-logind
  164 root        20   0  1792300   15340   12552 S   0.0   0.1   0:00.13 wsl-pro-service
  191 syslog      20   0  220548    5928    4656 S   0.0   0.0   0:00.10 rsyslogd
  220 _chrony      20   0   24364   11844    7144 S   0.0   0.1   0:00.10 chronyd
  229 _chrony      20   0   12092    2440    1796 S   0.0   0.0   0:00.01 chronyd
  255 root        20   0    5492    2732    2544 S   0.0   0.0   0:00.01 agetty
  264 root        20   0  123456   32504   17900 S   0.0   0.2   0:00.20 unattended-upgr
  332 root        20   0    3184    1104     980 S   0.0   0.0   0:00.00 SessionLeader
  333 root        20   0    3200    1244    1108 S   0.0   0.0   0:00.01 Relay(334)
  334 gayathri    20   0    6268    5600    3928 S   0.0   0.0   0:00.10 bash
  335 root        20   0    8648    5560    4716 S   0.0   0.0   0:00.02 login
  383 gayathri    20   0   22756   13336   10868 S   0.0   0.1   0:00.21 systemd
  385 gayathri    20   0   22912   4084    2080 S   0.0   0.0   0:00.00 (sd-pam)
  413 gayathri    20   0    6136    5484    3928 S   0.0   0.0   0:00.06 bash
  547 gayathri    20   0    8708    5272    4724 S   0.0   0.0   0:00.04 dbus-daemon
  551 gayathri    20   0   380960    8020    7372 S   0.0   0.0   0:00.01 at-spi-bus-laun
  558 gayathri    20   0    8540    5040    4676 S   0.0   0.0   0:00.00 dbus-daemon
  559 gayathri    20   0   476864   17644   14752 S   0.0   0.1   0:00.08 xdg-desktop-por
```